SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I Latisha. S (192524071) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Latisha.

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Cloud Auto-scaling and Load balancing..

Auto-scaling automatically increases or decreases cloud resources based on demand. Load balancing distributes incoming requests across multiple servers.

- Auto-scaling: Adds servers during traffic spikes and removes them when demand decreases.
- Load balancing: Distributes user requests evenly to prevent server overload.
- Reduced downtime: Health checks detect failed servers and redirect traffic to healthy servers.
- Monitoring tools: Track CPU usage, response time, traffic and errors to trigger scaling actions.
- Real world-example: During festival sale, an e-commerce website can add servers automatically to handle thousands of customers shopping simultaneously.

(*). Auto-scaling and load balancing improve availability, reduce downtime and provide a smooth user experience while controlling costs.

2. Secure Cloud Backup storage.

Cloud storage provides scalable and secure storage for large volumes of backup data.

⇒ Durability and redundancy: Cloud providers store multiple copies of data across devices or locations to reduce the risk of data loss.

⇒ Encryption : Protects backup data at rest and during transmission.

⇒ Access control: IAM and role-based permissions ensure that only authorized users can access backups.

⇒ cost advantages : Pay-as-you-go pricing and low-cost archival storage reduce hardware, maintenance and infrastructure expenses.

⇒ Disaster recovery example: If a company's local server fails due to a fire, it can restore its applications and files from cloud backups.

(*). Cloud backups provides reliable data protection, security, scalability and cost-effective disaster recovery.

3. Content Delivery Networks (CDNs) and global availability.

A content delivery networks (CDNs) is a group of distributed servers that delivers content from locations closer to users

* Improved performance : CDNs cache images, videos and other content near users, reducing loading time

* Reduced latency : Edge servers deliver content without requiring every request to travel to the main server.

* Global availability: Cloud providers offer multiple regions and availability zones to improve service availability.

* Reliability : Traffic can be redirected when a server or location fails.

Real world Example: A global video streaming application uses a CDN to deliver videos quickly to users all over the globe.

(P). CDNs and edge servers reduce latency, improve response time and provide a better experience for global users.

4. Cloud security and Regulatory compliance in Finance.

Financial institutions must protect sensitive customer information and follow applicable regulations.

→ Regulatory compliance: Cloud providers offer security controls and compliance - supporting services to help meet relevant standards and regulations.

→ Logging and auditing: Record user activities, transactions and changes to identify suspicious actions.

→ Monitoring: Detect unusual access patterns and possible security threats.

→ Encryption: Protect financial data at rest and in transit.

→ Example: A bank encrypts online payment information, restricts access to authorized employees and audits transactions to detect fraud.

(P). Encryption, access control, monitoring and auditing help financial institutions protect data and meet compliance requirements.

5. Containerization and Kubernetes :

Containers package applications with their dependencies, while VMs run separate guest operating systems

Feature	Containers	Virtual Machines.
Operating System	Share the host OS Kernel	Each VM runs a guest OS.
Resource usage	Generaly lower	Generally higher
Startup	Usually faster	Usually slower.
Portability	Portable across compatible environments	Portable across compatible hypervisors.

Roles of Kubernetes:

* Deployment = Automates application deployment
* Scaling = Adjusts containers based on demand
* Load balancing = Distributes traffic
* Self-healing = Restarts failed containers.

Example: An online shopping application uses separate containers for login, payments and orders. Kubernetes manages and scales them idenpendently